



Review article

Mortality, morbidity and healthcare costs of short-term high temperatures and heatwaves exposure in older populations: a global systematic review and meta-analysis

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ABSTRACT

Background: Exposure to heat extremes is associated with higher rates of mortality, morbidity and healthcare costs. In the context of population ageing and climate change, which is expected to make extreme heat events more intense, frequent, prolonged, and widespread, understanding their health impacts on older adults is crucial for informing targeted interventions.

Methods: This systematic review and meta-analysis searched PubMed, Embase, Web of Science, Scopus, CINAHL, and AgeLine for studies published between January 1, 1990, and March 15, 2025. Original research articles assessing morbidity, mortality, and healthcare costs among older adults during periods of high temperatures or heatwaves were included. Eligible studies used observational designs, such as time series, case-crossover, or case time-series, comparing risks across different exposures or time periods. Reviews, commentaries, grey literature, animal studies, and studies not focused on temperature as the primary exposure were excluded. Risk estimates were extracted from published studies. A random-effects meta-analysis was conducted to pool relative risks of morbidity and mortality, focusing on specific disease categories according to ICD-10 codes, climate zones, and demographic factors. The certainty and strength of the evidence were assessed following the Navigation Guide systematic review methodology framework. The study protocol was registered with PROSPERO (CRD42023423311).

Results: Out of 21,610 identified studies, 623 were eligible, spanning six continents and 305 locations classified by Köppen climate zones. Of these, 578 were included in the meta-analysis, 43 in a supplementary meta-analysis, and four on healthcare costs were narratively synthesised. A 1°C temperature rise was associated with higher morbidity (RR 1.016 [CI95% 1.012–1.019]) and mortality (RR 1.022 [CI95% 1.020–1.024]) among older people. The highest morbidity risks were linked to heat-related illnesses, external causes and endocrine diseases, and in tropical climates. Mortality risk was highest for intentional self-harm and external causes, chronic obstructive pulmonary disease (COPD), and in Mediterranean climates, with the strongest overall impact on those aged ≥ 85 years. Heatwaves increased morbidity (RR 1.189 [CI95% 1.146–1.233]), particularly for heat-related illnesses and external causes, with the highest overall risk seen in adults aged 75–84 and in subtropical climates. Mortality also rose (RR 1.117 [CI95% 1.105–1.129]), especially for cerebrovascular, endocrine and respiratory diseases, with the greatest overall risks found in tropical climates and among individuals aged ≥ 85 years. Despite the limited number of studies on healthcare costs they showed significant increases in costs during heat exposure for emergency department presentations, respiratory and ureter calculus hospitalisations. There was high heterogeneity with morbidity (high temperatures: $I^2 = 84.1\%$; heatwaves: $I^2 = 97.6\%$), whilst for mortality, there was

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no observed heterogeneity with high temperatures ($I^2 = 0.0\%$) and moderate heterogeneity with heatwaves ($I^2 = 72.2\%$). The strength of evidence was rated as “sufficient” for an association between heat exposure and increased mortality and morbidity in older adults, and “limited” for healthcare costs.

Conclusion: This review highlights the critical impact of heat exposure on the health of older adults, particularly in different climate zones, and indicates a potential increase in healthcare costs associated with this risk. With the global rise in temperatures and ageing populations, heat-related health risks will continue to increase. Effective, evidence-based interventions, such as tailored heat health action plans that consider individual risk factors, local resources, climatic characteristics and healthcare infrastructure, are crucial to minimise these risks.

1. Introduction

The anticipated increase in chronic and acute heat exposure among older adults poses a significant global public health challenge (Falchetti et al., 2024, Liu et al., 2022, Bunker et al., 2016, Liu et al., 2021b, Lee and Kim, 2016). Extreme heat increases mortality and morbidity, particularly in older individuals (Bunker et al., 2016, Faurie et al., 2022, Liu et al., 2021b, Liu et al., 2022, Lee and Kim, 2016, Cissé et al., 2022). It also imposes a significant economic burden on healthcare systems, with most healthcare costs attributed to older people (Wondmagegn et al., 2019). Older adults account for a large proportion of overall healthcare costs due to their higher prevalence of chronic diseases and healthcare utilisation (Kallestrup-Lamb et al., 2024, OECD 2023, WHO 2025). Age-related comorbidities, medication use, compromised physiological function, and financial constraints place many older adults at greater risk of the negative health effects of heat exposure (Zhang et al., 2016).

The global population aged 65 years and older is projected to increase from 761 million in 2021 to about 1.6 billion by 2050 (United Nations, 2023). The Global Burden of Disease Study in 2019, identified cardiovascular diseases, neoplasms, and chronic respiratory diseases, with ischemic heart disease, stroke, COPD, dementia, and lower respiratory infections as top causes of death among those aged 70 and older (GBD 2019 Ageing Collaborators, 2022). These climatic and demographic changes will likely lead to increases in both acute and chronic heat-sensitive diseases, presenting significant challenges to healthcare systems (Bunker et al., 2016, United Nations, 2023, Liu et al., 2021a, Liu et al., 2021b, Liu et al., 2022).

Although studies on healthcare costs related to heat exposure are limited, existing literature suggests the economic impact on the healthcare system is substantial (Wondmagegn et al., 2019, Wondmagegn et al., 2021, Lin et al., 2012, Yang et al., 2024, Wang et al., 2021).

A 2019 systematic review on heat-attributable healthcare costs encompassed ten studies, but only one specifically addressed healthcare costs among older adults (Wondmagegn et al., 2019). Heat-related healthcare expenses include ambulance services, emergency department visits, hospital admissions, medical devices, and diagnostic procedures (Wondmagegn et al., 2021, Wondmagegn et al., 2019, Lin et al., 2012). Estimating these costs offers an economic viewpoint on the burden of heat-related health effects (Wondmagegn et al., 2019).

Previous reviews have explored disease-specific heat-associated risks, such as cardiovascular diseases (Liu et al., 2022, Cheng et al., 2019), mental health conditions (Liu et al., 2021b, Byun et al., 2024), heat-related illnesses (Faurie et al., 2022), renal diseases (Liu et al., 2021a), diabetes (Song et al., 2021) and respiratory diseases (Cheng et al., 2019, Tran et al., 2025), finding older people to be at risk for both mortality and morbidity. However, few systematic reviews focused specifically on older people, quantifying heat-related morbidity and mortality outcomes. A 2016 meta-analysis showed that for every 1°C increase in temperature, mortality significantly increased for cardiovascular (3.44%), respiratory (3.60%), and cerebrovascular diseases (1.40%). Similarly, a 1°C increase in temperature was associated with higher morbidity for respiratory diseases (1.65%), diabetes (1.02%), genitourinary diseases (2.12%), and heat-related illnesses (14.83%) (Bunker et al., 2016). A 2011 meta-analysis indicated that individuals

aged 65 and older experienced a 2–5% increase in all-cause mortality for each 1°C rise in temperature (Yu et al., 2012). However, neither meta-analysis assessed the impact of heatwave exposure, explored climatic characteristics, nor conducted risk of bias (RoB), certainty, or strength of evidence assessments.

A more recent meta-analysis in 2024 specifically investigated mental and neurological outcomes in older adults and included both high temperature and heatwave exposures. It reported elevated risks for mental disorders during heat events and applied a RoB assessment (Byun et al., 2024). However, the analysis did not account for differences across climate zones and income classifications. While systematic reviews and meta-analyses have considered the impact of climate variability on these associations, the risk of heat-related mortality and morbidity across different climate zones remains unassessed. Other reviews synthesised previous findings narratively (Åström et al., 2011, Kaltsatou A, 2018). Moreover, many studies on the relationship between heat and health have been published since.

Appendix 1, Tables S2 and S3 provides a systematic identification and mapping of relevant reviews, assessing their scope, methodological characteristics, and the evidence gaps addressed by this review. No previous review has comprehensively synthesised mortality, morbidity, and healthcare cost outcomes in older populations, nor provided stratified analyses by climate zones or demographic subgroups.

This systematic review and meta-analysis addresses these gaps by quantifying the effects of heat exposure, including high temperatures and heatwaves, on mortality, morbidity, and healthcare costs among older adults based on available epidemiological evidence. It focuses on specific disease categories according to ICD-10, climate zones, and demographic factors, while also assessing risk of bias, overall certainty, and strength of the evidence.

2. Methods

This systematic review and meta-analysis followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Appendix 1, Table S1) (Page et al., 2021). The protocol for this review was registered with PROSPERO (CRD42023423311).

The primary research question guiding this systematic review and meta-analysis is framed using the PECO structure (Population, Exposure, Comparator, Outcomes) (Morgan et al., 2018) as follows:

Among older adults (P), is exposure to high ambient temperatures or heatwaves (E), associated with increased risk of mortality, morbidity, or healthcare costs (O), compared with cooler reference periods or non-heatwave days (C)?

2.1. Eligibility criteria

The inclusion and exclusion criteria for study selection were structured using the PECOS framework (Morgan et al., 2018), as outlined in Table 1 and explained in detail below.

1) Population: Although the United Nations defines older persons as those aged ≥ 60 years, no single age threshold is used consistently across research or policy. In some settings, lower thresholds reflect not only shorter life expectancy but also socio-cultural perceptions of

Table 1
Inclusion and exclusion criteria based on the PECOS Framework.

Element	Inclusion criteria	Exclusion criteria
Population	Older adults ≥ 60 years; any sex, ethnicity, or geographic location.	Populations < 60 years; non-human studies; populations limited to specific pre-existing conditions.
Exposure	High ambient temperatures and heatwaves, as defined by each study (e.g., maximum temperature, apparent temperature, heat index).	Studies where temperature was not the main exposure; studies focused only on cold or compared seasons (e.g., summer vs. winter) without analysing temperature or heatwaves as defined exposures; simulated, experimental, or indoor temperature exposures.
Comparator (s)/control:	Cooler reference periods, non-heatwave days, reference temperatures with minimum risk of morbidity or mortality.	Studies lacking a comparator or reference period.
Outcome(s):	All-cause and disease-specific mortality and morbidity outcomes (ICD-10 codes [A00–Z99], morbidity linked to healthcare utilisation; healthcare costs specified as costs related to healthcare utilisation during heat exposure.	Studies reporting only self-reported symptoms, subjective wellbeing, or functional performance tests.
Study Design:	Observational epidemiological studies using time-series, case-crossover, or case time-series designs. Peer-reviewed journal articles published in English.	Self-reported or questionnaire-based temperature exposure or health outcomes. Experimental, lab-based or simulation studies; reviews, editorials, conference abstracts, theses, or grey literature. Non-English publications.

ageing (United Nations, 2024). This review included studies using definitions of 60 years and above, as well as two studies from low- and lower-middle-income countries, such as Kenya (≥ 50 years) and Senegal (≥ 55 years), where lower thresholds aligned with local demographic and health contexts (United Nations, 2024, Bunker et al., 2016). These were included to avoid data loss, as evidence from these regions is limited. Studies were also included if they stratified their analysis for older adults.

- 2) Exposure: High ambient temperature and heatwave exposure included short-term exposure to outdoor heat. Eligible exposure metrics included e.g. minimum, maximum, mean, apparent temperature, heat index, excess heat factor (EHF), and other validated indicators. Exposure could be modelled as continuous (e.g., per 1°C increase above a reference) or categorical, using thresholds such as temperature percentiles compared to a reference (e.g., 29°C vs. 15°C), absolute cut-off values (e.g., $>35^\circ\text{C}$), or defined heatwave periods (e.g., ≥ 3 consecutive hot days).

Heatwaves were eligible if defined by the study as a period of unusually high temperatures, typically over consecutive days, or based on statistical models such as the EHF (Nairn and Fawcett, 2015). Threshold definitions varied by study but were required to reflect excess heat conditions compared to local climate norms.

We excluded studies that only adjusted for temperature as a covariate, lacked an appropriate comparator, or assessed only cold or compared seasons (e.g., summer vs. winter) without analysing temperature or heatwaves as defined exposures. Experimental or simulation-based studies, indoor exposures, and self-reported exposure assessments were also excluded.

- 3) Comparators: Studies that included cooler reference periods or populations not exposed to high temperature or heatwaves, such as

non-heatwave days, lower temperature percentiles, or baseline/reference temperatures defined by the study. Studies without an appropriate comparator or reference condition were excluded.

- 4) Outcomes: All-cause and disease-specific mortality and morbidity among older adults during periods of high temperature or heatwaves were included. Morbidity outcomes were eligible if they were linked to healthcare utilisation, such as hospital admissions, emergency department visits, ambulance call-outs, or outpatient visits. Healthcare cost outcomes were included when directly related to healthcare utilization during heat exposure. Disease classifications followed the ICD-10 system (codes A00–Z99) and encompassed heat-sensitive categories such as cardiovascular, respiratory, renal, genitourinary, metabolic, and heat-related illnesses. Studies were included if they reported outcomes on a daily resolution and provided effect estimates such as relative risk, odds ratio, percent change, or incidence rate ratios with a 95% confidence interval (CI) based on empirical epidemiological data from observed historical health and temperature records. We excluded studies reporting only self-reported symptoms, functional performance, or subjective wellbeing, as well as those focusing solely on physiological parameters or biomarkers without associated health outcomes. Studies using simulated, projected, or scenario-based data were excluded. Studies reporting indirect, productivity, or mortality costs (e.g., value-of-statistical-life estimations or economic valuations of years of life lost) were also excluded.
- 5) Study designs: Eligible studies used observational ecological study designs such as time series, case crossover, or case time-series designs to compare risks over different exposures or time periods. Only peer-reviewed, original studies in English were included. We excluded studies based on self-reported or questionnaire-based exposure or outcome assessment, as well as experimental, simulation-based, or laboratory studies. Also excluded were reviews, editorials, commentaries, theses, conference abstracts, and non-English publications.

2.2. Search strategy

A comprehensive literature search was conducted with guidance from the university librarian across six databases: PubMed, Embase, Web of Science, Scopus, CINAHL, and AgeLine. The search strategy was structured around the PECO framework and applied keywords and Boolean operators covering the following domains (full search terms listed in Appendix 1, Table S4):

Population (e.g., “Older people”, “Aged”), Exposure/Comparator (e.g., “Heat”, “Hot weather”, “High Temperature”), Outcomes (e.g., “Health outcomes”, “Morbidity”, “Mortality”, “Health care cost”).

Within each domain, search terms were combined using “OR” to include all relevant variants. These groups were then connected using the “AND” operator.

Pilot testing of the search strategy was conducted during its development. Based on preliminary findings, outcome-related terms were refined to include specific morbidity indicators such as hospitalisation, ambulance call-out, emergency department visits, and outpatient visits.

To assess the completeness and validity of the search, we reviewed the reference lists of 25 relevant systematic reviews identified through targeted searches (Appendix 1, pp. S3–S12). This step helped identify additional eligible primary studies not retrieved through the initial search and confirmed adequate conceptual coverage.

The final search was conducted for studies published between January 1, 1990, and May 31, 2023. An updated search was performed on March 15, 2025, to capture newly published literature.

2.3. Study selection and data extraction

All retrieved articles from the six databases were imported into EndNote (The EndNote Team, 2013), and duplicates were automatically

identified based on combinations of author names, publication year, article title, DOI, and journal information and each flagged record was manually reviewed and duplicates removed.

The screening of titles, abstracts and full texts was conducted in a two-stages: initial title and abstract screening to remove irrelevant records, followed by full-text review for eligibility. Screening was performed by four authors (SG, BMV, DB, JLi) through a double-blind independent screening process using Rayyan (Ouzzani et al., 2016). An overall inter-reviewer agreement rate of 96% was achieved, and discrepancies were resolved through a third reviewer. Eight authors (SG, BMV, DB, YL, MB, JLi, YH, YS) independently extracted and cross-checked the data from each study into a customised form. Any inconsistencies were resolved through discussion and cross-verification among reviewers. The extracted data included the following: author, year, location, study period, study design, exposure measurements, confounders, statistical models, effect estimates with corresponding 95% CIs.

All relevant effect estimates were collected from studies. Risk estimates were obtained from texts, tables, appendices or graphical data using WebPlotDigitizer (Rohatgi, 2017). Additionally, national income levels were recorded based on World Bank data (World Bank, 2024). Climate zones were identified using the Köppen–Geiger classification system (Peel et al., 2007).

2.4. Risk of bias assessment

The RoB for individual studies was assessed using a modified version of the Office of Health Assessment and Translation (OHAT) RoB Tool, following the Navigation Guide for systematic reviews in environmental health (OHAT, 2019; Liu et al., 2022; Johnson et al., 2014). Each study was evaluated across eight domains: exposure assessment, outcome assessment, confounding, selection bias, incomplete or missing data, selective reporting, conflict of interest, and other sources of bias. For each domain, risk was rated as “low,” “probably low,” “probably high,” or “high” risk based on pre-specified criteria (Appendix 1, Table S5). The intermediate categories (“probably low” and “probably high”) were applied when information was insufficient for a definitive judgment but provided indirect evidence toward lower or higher risk, respectively, consistent with Navigation Guide conventions. The overall RoB was determined based on three key domains: exposure, outcome, and confounding (Liu et al., 2022). The rationale for overall bias ratings is provided in Appendix 1, Table S6.

Six reviewers (SG, YL, MB, JLi, YH, YS) independently conducted RoB assessments and resolved disagreements through discussion.

2.5. Synthesis of the evidence

Study findings were synthesised both quantitatively and narratively. A DerSimonian and Laird random-effects *meta*-analysis was conducted (DerSimonian and Laird, 1986) to pool effect estimates from studies that provided relative risk (RR), odds ratio (OR), incidence rate ratio (IRR), or other effect estimates that could be converted to RR (e.g. percentage change or excess risk) (DerSimonian and Laird, 1986).

For high temperature exposures, estimates were converted to RR per 1°C increase above the study-specific reference temperature when studies reported temperature contrasts or increments other than 1°C (Liu et al., 2022). This standardisation/conversion assumed a log-linear relationship within the high-temperature range and was used as a statistical scaling to improve comparability across studies, recognising that a 1°C increase in different metrics does not necessarily represent the same underlying physical change in thermal exposure.

Separate *meta*-analyses were performed for heatwave studies, pooling RRs of outcomes during heatwave versus non-heatwave periods. Heatwave definitions varied widely across studies in terms of temperature thresholds and durations. To account for this variability, subgroup analyses were performed by heatwave intensity – low (90th–93rd

percentile), middle (94th–96th percentile), and high (97th–99th percentile) (Liu et al., 2022) – as well as durations (e.g. ≥ 2 days, ≥ 3 days, ≥ 4 days).

Studies reporting high temperature exposure using categories or percentiles only (e.g. 95th vs. 50th percentile, or 99th (34°C) vs. reference temperature) without reporting the corresponding absolute temperature were included in a secondary *meta*-analysis (Appendix 1, Tables S16 and S17) in which effect estimates were pooled as reported.

Subgroup analyses were performed when four or more estimates were available to explore variations in disease outcomes, demographic factors, national income levels, Köppen–Geiger climate zones (Peel et al., 2007), continents and healthcare utilisation outcomes. Disease outcomes were categorised using the International Classification of Diseases, Tenth Edition (ICD-10) nomenclature and hierarchy.

For the studies that reported multiple estimates for either individual lag days (e.g. lag 0, 1, 2) or age groups (e.g. 60–85, ≥ 85 years) without providing an overall estimate, these estimates were combined into a single study-specific effect using an inverse-variance fixed-effect *meta*-analysis. Effect estimates were first transformed to the log scale and weighted by the inverse of their variance to obtain one overall estimate per study for inclusion in the main analysis (Liu et al., 2022; Liu et al., 2021b; Byun et al., 2024; Higgins et al., 2024). Multi-location studies included estimates from all locations for pooled effects (Liu et al., 2022).

For outcomes that could not be statistically pooled (i.e., health care costs), a narrative synthesis was performed. This included a structured summary of the direction, magnitude, and precision of reported effect estimates, in line with recommendations (Popay et al., 2005).

Heterogeneity was assessed using Cochran's Q and Higgins' I² statistics, with I² values categorised as low (0–25%), moderate (26–74%), and high ($\geq 75\%$) (Higgins et al., 2003). Additionally, 80% prediction intervals (PI) were used to estimate the range within which the true RR for future studies is likely to fall, with severe heterogeneity indicated if the 80% PI included the null effect and was more than twice the width of the 95% CIs (Int'Hout et al., 2016).

Publication bias was evaluated using Egger's test, funnel plots (Egger et al., 1997), and the trim-and-fill method (Duval and Tweedie, 2004). A univariate *meta*-regression analysis was performed to explore study-level characteristics that may explain between-study heterogeneity in effect sizes (Liu et al., 2022; Bown and Sutton, 2010; Deeks et al., 2024). The variables tested included Köppen–Geiger climate group, lag effects, age group, sex, continent, hemisphere, heatwave intensity and duration, and income classification.

Sensitivity analyses were undertaken by categorising studies according to temperature metrics, study design, lag periods, overall RoB rating, seasonal variations (warm season vs. whole year), air pollution and humidity adjustment, as well as modelling approach (non-linear or linear) (Liu et al., 2022). Leave-one-out analyses were used to assess the impact of individual studies on pooled RRs.

Statistical analyses were carried out using R software (version 4.4.1) (R Core Team, 2024) with the package metafor (Viechtbauer, 2010).

2.6. Certainty of the evidence

The certainty of the evidence was assessed following the Navigation Guide for systematic reviews of observational studies in environmental health (Johnson et al., 2014; Johnson et al., 2016). This process involved three steps, with the first step—Risk of Bias (RoB) assessment—described in Section 2.4.

2) The certainty of evidence across studies was assessed using the Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) framework (Guyatt et al., 2011). Evidence for each exposure–outcome pair was initially rated as “moderate” and could be downgraded for domains such as risk of bias, inconsistency, indirectness, imprecision, and publication bias. It could be upgraded based on evidence for a dose–response relationship, large magnitude of effect, or residual confounding that would likely attenuate the observed

association. The final certainty rating was assigned as “high”, “moderate”, or “low”, based on the net impact of downgrading and upgrading (Johnson et al., 2014, Johnson et al., 2016). The decision rules and justifications for these adjustments are provided in Appendix 1, Table S7, which are also described in detail elsewhere (Johnson et al., 2014, Johnson et al., 2016).

3) The overall strength of the evidence was determined by combining certainty assessments with consistency of findings and confidence in estimated effects. Evidence was categorized as “sufficient,” “limited,” or “inadequate,” following the Navigation Guide framework (Johnson et al., 2014, Johnson et al., 2016).

3. Results

3.1. Study selection and characteristics

The database searches retrieved 21,610 records. After filtering non-English, non-human or non-journal articles and removing duplicates 12,131 records remained for title and abstract screening. Most records were excluded at this stage as they were unrelated to heat exposure and health outcomes, were the wrong publication type or not based on human data. Hence, 812 studies were assessed in full text. A further 210 studies were excluded, primarily due to irrelevant outcomes or exposures, unsuitable study design, or populations outside the inclusion criteria (see Fig. 1 for details). An additional 21 studies were identified through the screening of 25 relevant systematic reviews, which had been retrieved via database searches and included data on older adults, suggesting the primary search was comprehensive. In total, 623 studies met the inclusion criteria (see Fig. 1).

Table 2 provides an overview of the 623 included studies, categorised by exposure type and reported outcome.

A random-effects *meta*-analysis was conducted for morbidity and mortality outcomes in relation to high temperature and heatwave exposure (n = 578). An additional 43 studies, which reported high temperature effects using non-standardizable exposure metrics (e.g. high temperature vs. study-defined reference such as minimum risk or percentile-based values), were synthesised in a supplementary *meta*-

Table 2

Overview of exposure and outcome from included studies.

Exposure	Outcome	Number of Studies
High Temperatures	Mortality	221
	Morbidity	224
	Health Care Costs	3
Heatwaves	Mortality	95
	Morbidity	73
	Health Care Costs	1

Note: The same study may appear in multiple outcome categories if it reported on more than one health and exposure outcome.

analysis (Appendix 1, Tables S16 and S17).

A *meta-analysis* for healthcare costs was not feasible due to the limited number of eligible studies. Four studies reported healthcare costs; two of them also contributed to the *meta-analysis* as they reported morbidity and heat exposure outcomes alongside cost estimates, while two were included in the narrative synthesis only. Key findings from all included studies are summarised in the [supplementary material](#) (Appendix 2).

The included studies were published between 1995 and 2025 and were conducted across Asia, Africa, Australia, Europe, North America, and South America. There were 154 location-specific studies for mortality and 151 for morbidity, which could be classified by the Köppen-Geiger climate zones: A (tropical), B (arid), C (temperate—including Mediterranean, Oceanic, and subtropical climates), D (continental), and E (polar) (Peel et al., 2007). Most studies were conducted in temperate (C) and continental (D) climates, primarily across North America, Europe, and East Asia (Fig. 2). A complete list of all study countries and locations is provided in Appendix 1, Table S8.

3.2. Risk of bias

The risk of bias assessment was low for 159 studies (27.4%), probably low for 380 studies (65.5%), probably high for 40 studies (6.9%), and high for 1 study (0.2%).

Studies were assessed across eight bias domains using a modified

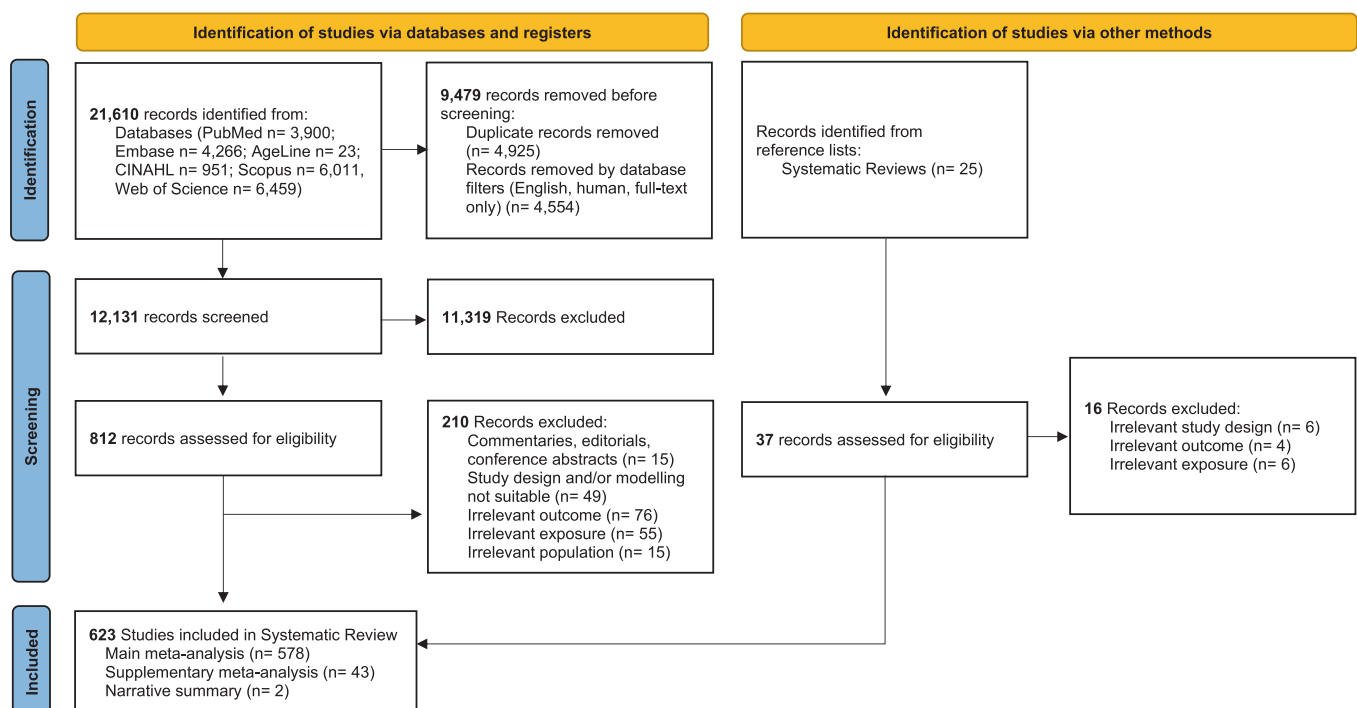


Fig. 1. PRISMA flow diagram of literature search and study inclusion.

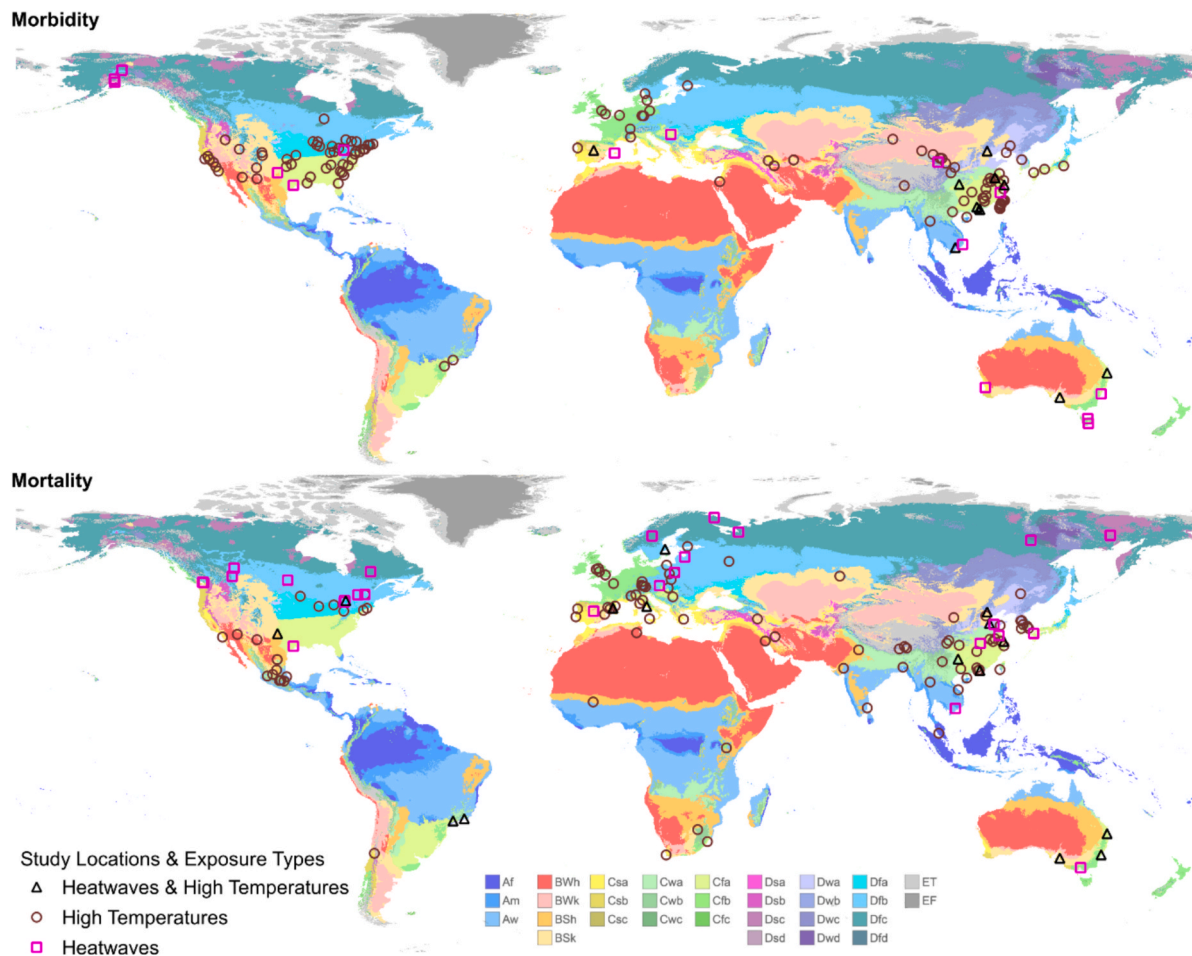


Fig. 2. Distribution of the 305 location-specific studies by exposure and outcome (a: Morbidity, b: Mortality) across the Köppen-Geiger climate zones. The map was created using a high-resolution (1 km*1km) Köppen-Geiger climate classification map (Beck et al., 2023).

OHAT tool, as described in Section 2.4. Overall, most studies showed low or probably low RoB in outcome assessment, selection bias, missing data, selective reporting, conflict of interest, and other sources of bias. However, higher RoB was frequently observed for exposure assessment and confounding bias, often due to the use of single monitoring stations for a large geographical area, lack of population weighting, or inadequate adjustment for critical time-varying confounders such as seasonality or humidity.

This pattern held across most exposure–outcome categories, with the majority of studies rated as low or probably low overall RoB, and only a small proportion judged as probably high or high. Summary plots were created using the Risk of Bias Visualization Tool (ROBVIS) to illustrate the percentage of RoB ratings across domains for each exposure–outcome combination (Fig. 3) (McGuinness and Higgins, 2020). The full RoB for each individual study, along with the assessment criteria, are presented in (Appendix 3).

3.3. Synthesis of high temperature effects

Overall, a 1°C rise in temperature increased morbidity (RR 1.016 [CI95% 1.012–1.019]) (Fig. 4) and mortality (RR 1.022 [CI95% 1.020–1.024]) among older individuals (Fig. 5). The highest morbidity risks were linked to heat-related illnesses (RR 1.228 [CI95% 1.155–1.307]), external causes (RR 1.109 [CI95% 1.071–1.149]), and endocrine diseases (RR 1.035 [CI95% 1.010–1.059]), while cardiovascular, digestive, infectious, and mental diseases showed no significant associations. Individuals aged ≥ 85 years, those in tropical climates, lower-middle-income countries, and residents of Asia experienced the

largest increases in morbidity per 1°C rise (Fig. 4).

Mortality risk was mostly elevated for intentional self-harm (RR 1.068 [CI95% 1.026–1.113]), external causes (RR 1.039 [CI95% 1.016–1.063]), COPD (RR 1.038 [CI95% 1.001–1.076]), cerebrovascular (RR 1.031 [CI95% 1.012–1.049]), respiratory (RR 1.029 [CI95% 1.019–1.039]) and cardiovascular diseases (RR 1.021 [CI95% 1.014–1.027]) (Fig. 5).

The overall test of heterogeneity was statistically significant for continent (QM (df = 5) = 24.8609, $p = 0.0001$, $I^2 = 0.0\%$), age group (QM (df = 2) = 11.7780, $p = 0.0028$, $I^2 = 0.0\%$), and hemisphere (QM (df = 1) = 11.3437, $p = 0.0008$), suggesting that these factors may significantly modify the association between high temperatures and mortality. Among continents, Australia showed the strongest increase in mortality (RR 1.075 [95% CI 1.052–1.098], $p < 0.0001$), followed by Europe (RR 1.019 [95% CI 1.012–1.027], $p < 0.0001$) Asia (RR 1.010 [95% CI 1.002–1.018], $p = 0.0129$), and North America (RR 1.009 [95% CI 1.002–1.015], $p = 0.0069$).

Age-stratified effects showed significantly increased mortality for the reference group aged 60–74 years (RR 1.015 [95% CI 1.009–1.022], $p < 0.0001$) and those aged ≥ 85 years (RR 1.018 [95% CI 1.008–1.029], $p = 0.0006$).

Hemispheres showed significantly elevated mortality risks. The Northern Hemisphere (reference) had an effect estimate of RR 1.021 [95% CI 1.018–1.024], $p < 0.0001$, while the Southern Hemisphere showed a comparable effect (RR 1.021 [95% CI 1.009–1.033], $p = 0.0008$) (Appendix 1, Table S9).

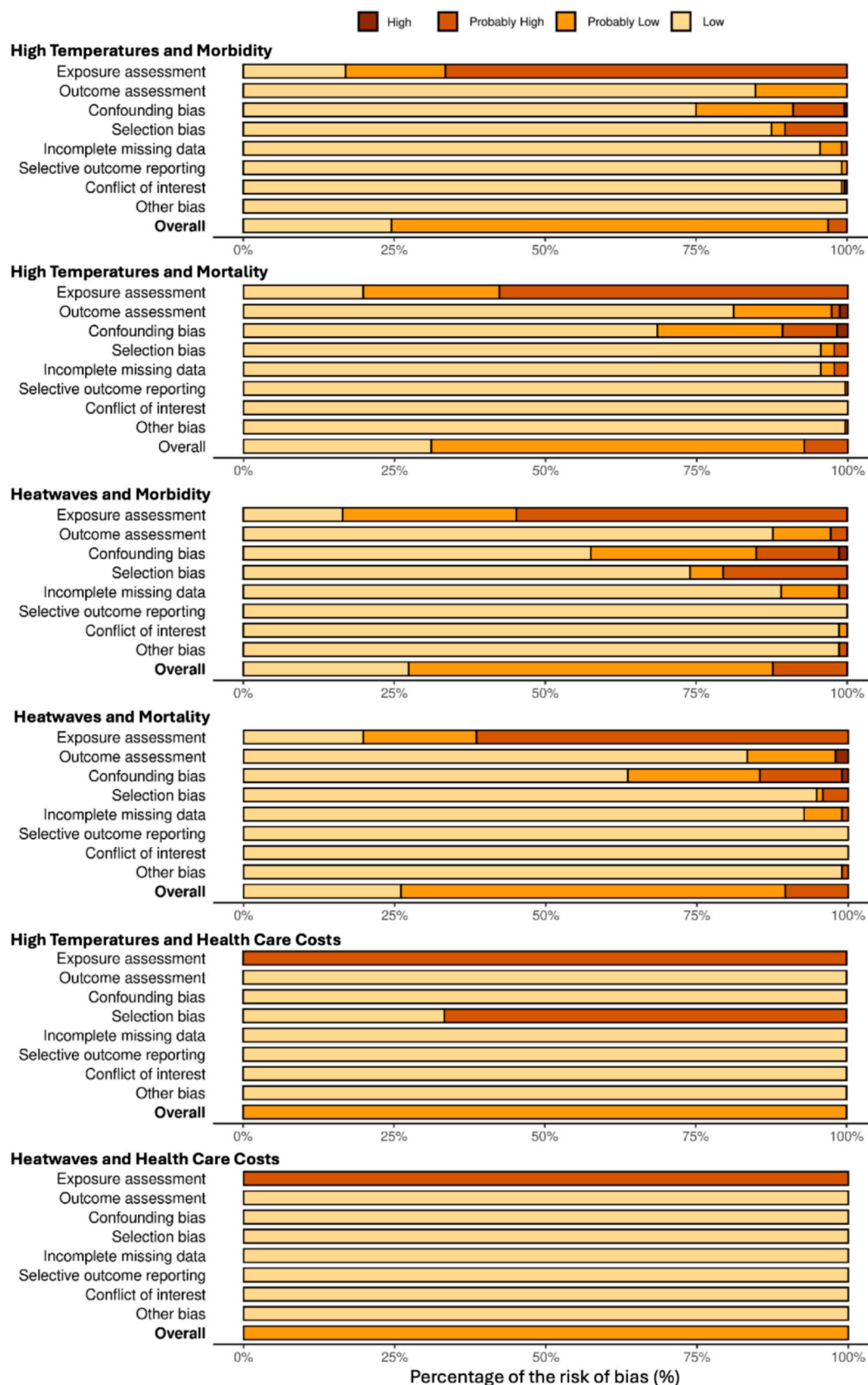


Fig. 3. Percentage distribution of Risk of Bias ratings across each bias domain for each exposure-outcome combination.

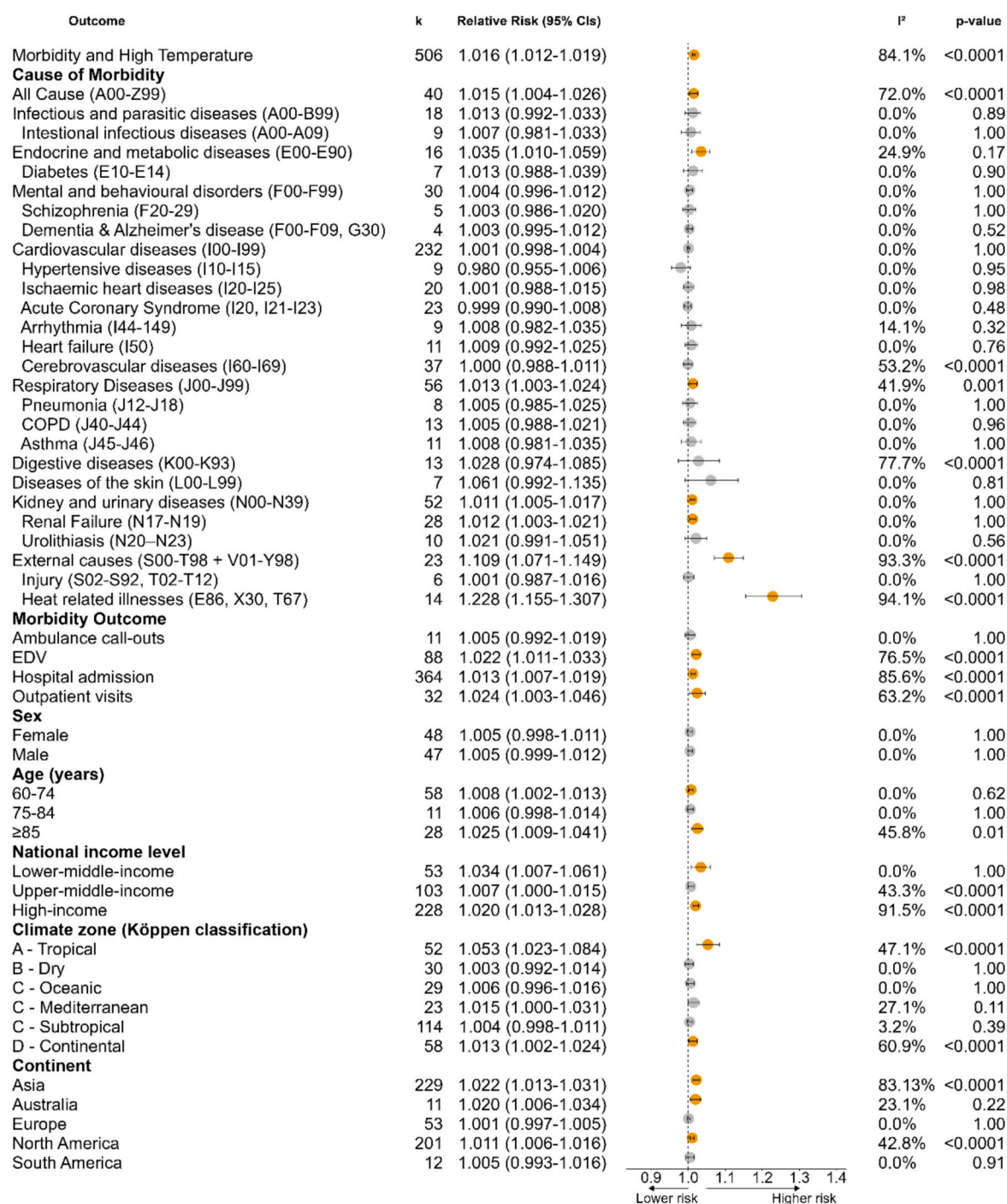


Fig. 4. Random-effects *meta*-analysis results showing change in RR and 95% CIs per 1°C increase in temperature for morbidity among older adults in different groups. k = the number of effects estimates; 95% CIs = 95% Confidence interval, I² = heterogeneity statistic; p-value = statistical significance of heterogeneity. Yellow markers indicate statistically significant associations; grey markers indicate non-significant associations.

3.4. Synthesis of heatwave effects

Heatwaves significantly amplified morbidity (RR 1.189 [CI95% 1.146–1.233]), with the most substantial increases during moderate-intensity heatwaves (RR 1.229 [CI95% 1.078–1.401]) and heatwaves lasting for ≥ 2 days (RR 1.212 [CI95% 1.137–1.292]) (Fig. 6). The largest increases were observed for external causes (RR 2.371 [CI95% 1.764–3.187]), and heat-related illnesses (RR 3.693 [CI95% 2.361–5.778]). High risks were also noted for endocrine diseases (RR 1.176 [CI95% 1.077–1.285]), fluid and electrolyte disorders (RR 1.155 [CI95% 1.057–1.262]) and renal failure (RR 1.146 [CI95%

1.041–1.262]). Subtropical climates, Australia, and high-income countries experienced the greatest morbidity risks. (See Fig. 6)

Meta-regression identified age group as a significant risk factor (QM (df = 2) = 7.9391, p = 0.0189, I² = 94.82%). The highest morbidity risk was observed in the 75–84 age group (RR 1.513 [95% CI 1.103–2.074], p = 0.0101), compared to the reference group (60–74 years). No significant difference was found for individuals aged ≥ 85 years (RR 0.953 [95% CI 0.792–1.147], p = 0.6106). However, the limited number of studies (n = 7) for the 75–84 group may affect reliability. Heatwave duration was also a significant moderator (QM (df = 3) = 15.4454, p = 0.0015, I² = 91.53%), although longer events (≥3–≥4 days) did not

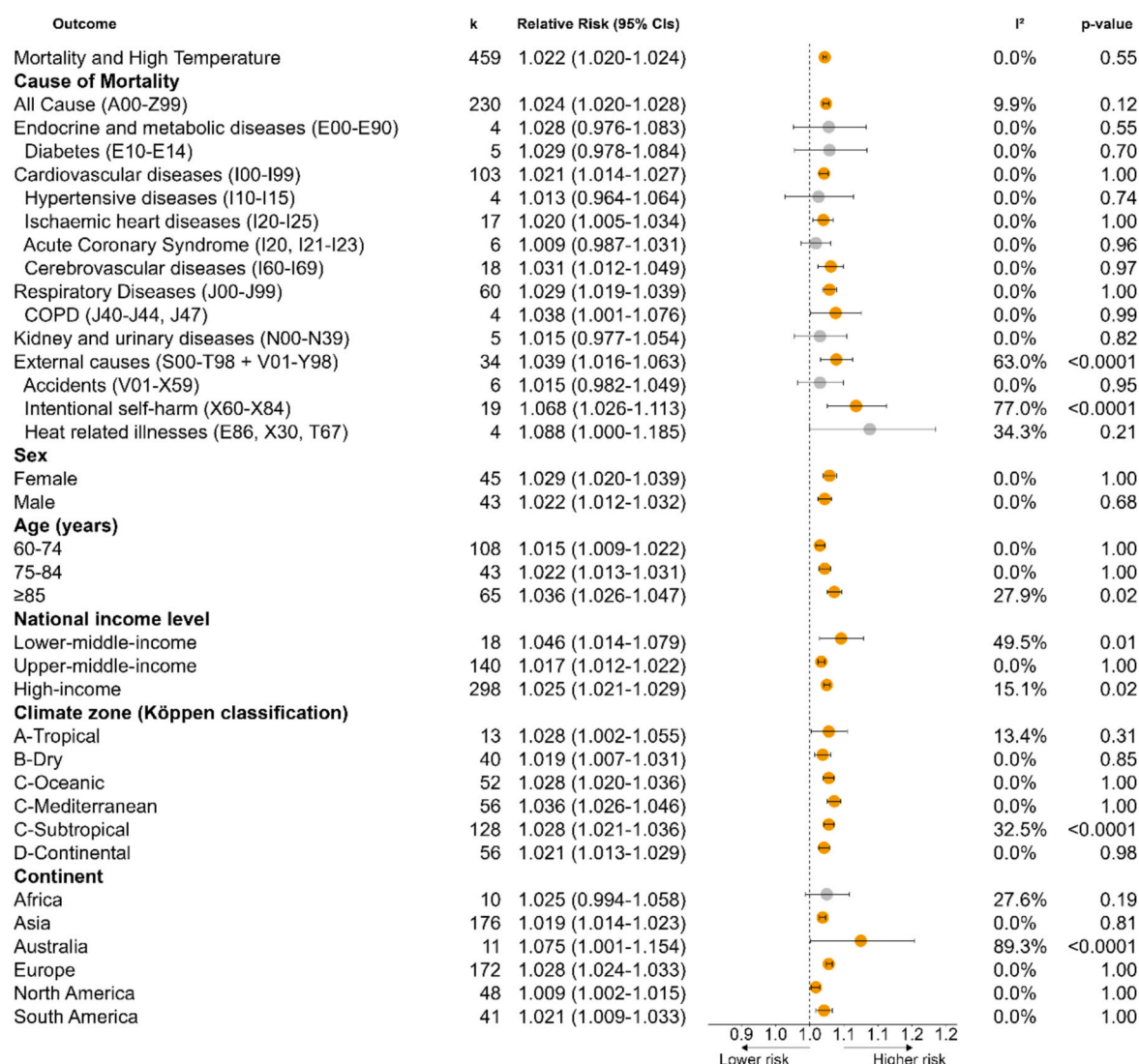


Fig. 5. Random-effects meta-analysis results showing change in RR and 95% CIs per 1°C increase in temperature for mortality among older adults in different groups. k = the number of effects estimates; 95% CIs = 95% Confidence interval, I² = heterogeneity statistic; p-value = statistical significance of heterogeneity. Yellow markers indicate statistically significant associations; grey markers indicate non-significant associations.

show additional risk compared with shorter (≥ 2 -day) heatwaves. Hemisphere significantly explained variation in morbidity effect sizes across studies (QM (df = 1) 7.4009, $p = 0.0065$, $I^2 = 97.56\%$), with higher morbidity observed in the Southern Hemisphere (RR 1.175 [95% CI 1.046–1.320], $p = 0.0065$) (Appendix 1, Table S10).

Mortality rose (RR 1.117 [CI95% 1.105–1.129]) during heatwaves, with the highest increases occurring during high-intensity heatwaves (RR 1.165 [CI95% 1.132–1.200]) and heatwaves lasting for ≥ 5 days (RR 1.309 [CI95% 1.195–1.434]). Strong risks were noted for cerebrovascular (RR 1.295 [CI95% 1.152–1.455]), respiratory diseases (RR 1.146 [CI95% 1.082–1.214]) and endocrine diseases (RR 1.149 [CI95% 1.067–1.238]). Elevated risks were also noted for diabetes, cardiovascular and ischaemic heart diseases. Europe and tropical climates showed the highest mortality risks (See Fig. 7).

Meta-regression showed significant effect modification by age group (QM (df = 2) = 38.0204, $p < 0.0001$, $I^2 = 32.09\%$), continent (QM (df = 5) = 22.3333, $p = 0.0005$, $I^2 = 70.46\%$), and income classification (QM (df = 2) 24.2293, $p < 0.0001$, $I^2 = 72.57\%$). Mortality risk was highest in adults ≥ 85 years (RR 1.182 [95% CI 1.120–1.247], $p < 0.0001$), followed by the reference group aged 60–74 years (RR 1.082 [95% CI 1.053–1.111], $p < 0.0001$) and those aged 75–84 years (RR 1.080 [95% CI 1.010–1.155], $p = 0.0243$). Across continents, the largest effect was

observed in Africa (RR 1.461 [95% CI 1.130–1.889], $p = 0.0038$), followed by Europe (RR 1.100 [95% CI 1.041–1.161], $p = 0.0006$) and Asia (RR 1.066 [95% CI 1.017–1.118], $p = 0.0076$) compared with North America (reference: RR 1.057 [95% CI 1.017–1.099], $p = 0.0052$). High-income countries showed a significant association with heatwave-related mortality (RR 1.079 [95% CI 1.056–1.103], $p < 0.0001$), as did upper-middle-income countries (RR 1.098 [95% CI 1.058–1.140], $p < 0.0001$). Heatwave duration was also a significant moderator (QM (df = 3) = 15.2109, $p = 0.0016$, $I^2 = 78.61\%$); longer events (≥ 4 – ≥ 5 days) were associated with higher mortality (Appendix 1, Table S10).

3.5. Synthesis of healthcare cost outcomes

Across the four identified studies (Table 3), two were conducted in high-income countries, the United States (Lin et al., 2012) and Australia (Wondmagegn et al., 2021), and two in an upper-middle-income country, China (Yang et al., 2024, Wang et al., 2021)

Wondmagegn et al. (2021) was the only study that assessed heatwaves, using the Excess Heat Factor, and reported significant increases in emergency department visits and large attributable costs among adults aged ≥ 65 years in Adelaide, with higher costs during severe events.

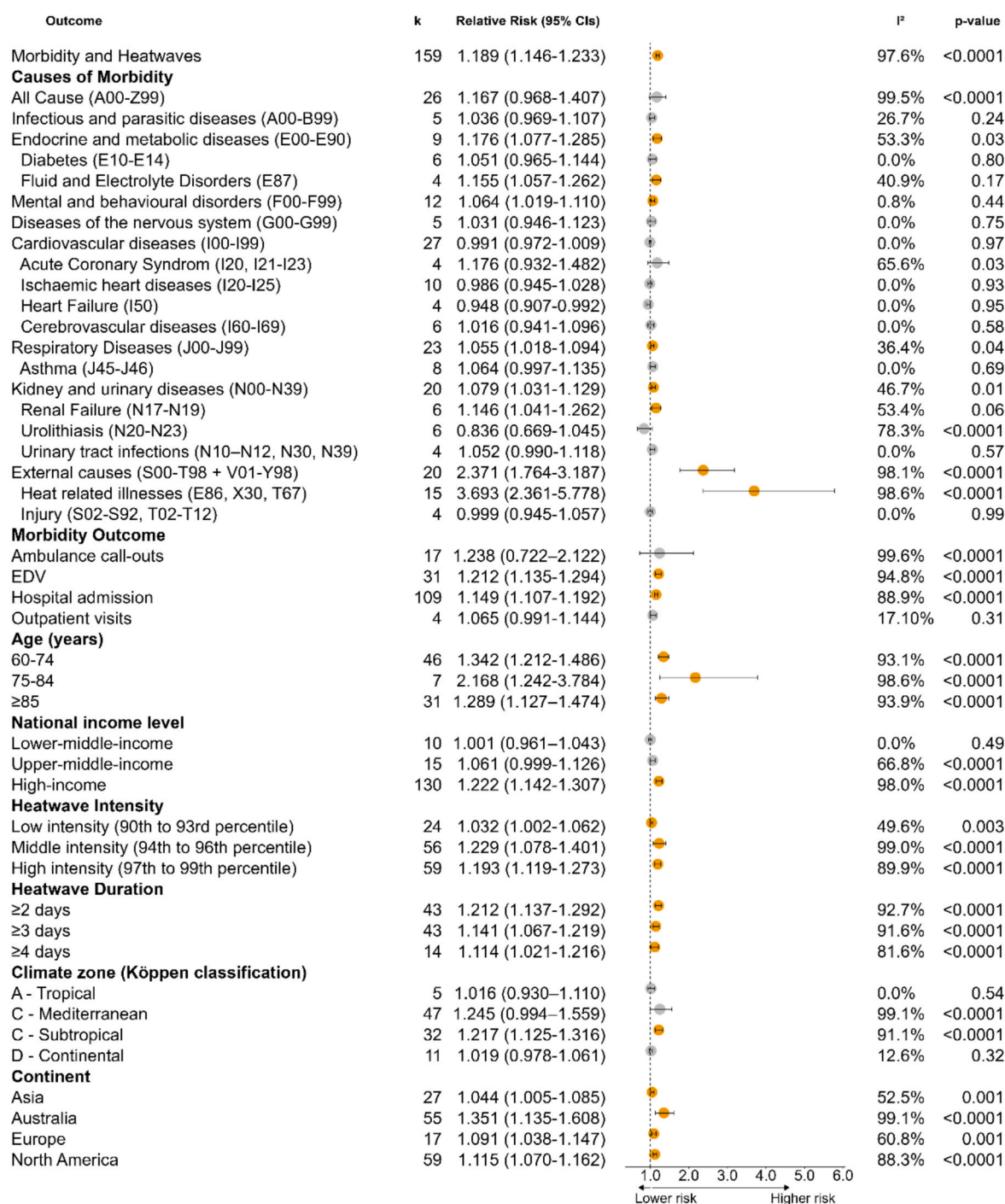


Fig. 6. Random-effects *meta*-analysis results showing change in RR and 95% CIs during heatwaves for morbidity among older adults in different groups. k = the number of effects estimates; 95% CIs = 95% Confidence interval, I² = heterogeneity statistic; p-value = statistical significance of heterogeneity. Yellow markers indicate statistically significant associations; grey markers indicate non-significant associations.

The remaining three studies applied continuous temperature metrics to examine incremental cost increases per degree or percentile change. Lin et al. (2012) found significant increases in respiratory hospitalisations and inpatient costs with higher apparent temperature in New York State, with greater excess costs in those aged ≥ 75 years compared with 65–74 years. (Yang et al., 2024) reported higher daily medical costs for ureteral calculi with increasing mean temperature in Nanjing.

Wang et al. (2021), analysing 17 sites across China, found small variations in relative risks across age groups and cost types for emergency department visits. Examination costs were slightly elevated among adults aged 60–74 years but lower among those ≥ 75 years, while

medication costs showed the opposite pattern, with small increases observed in the older group.

Overall, heat exposure was generally associated with higher healthcare costs among older adults, although the magnitude and direction of effects varied by exposure metric, health service type, and age group.

3.6. Certainty of evidence

Following the Navigation Guide guidelines for observational studies (Johnson et al., 2014), an initial “moderate” certainty rating was

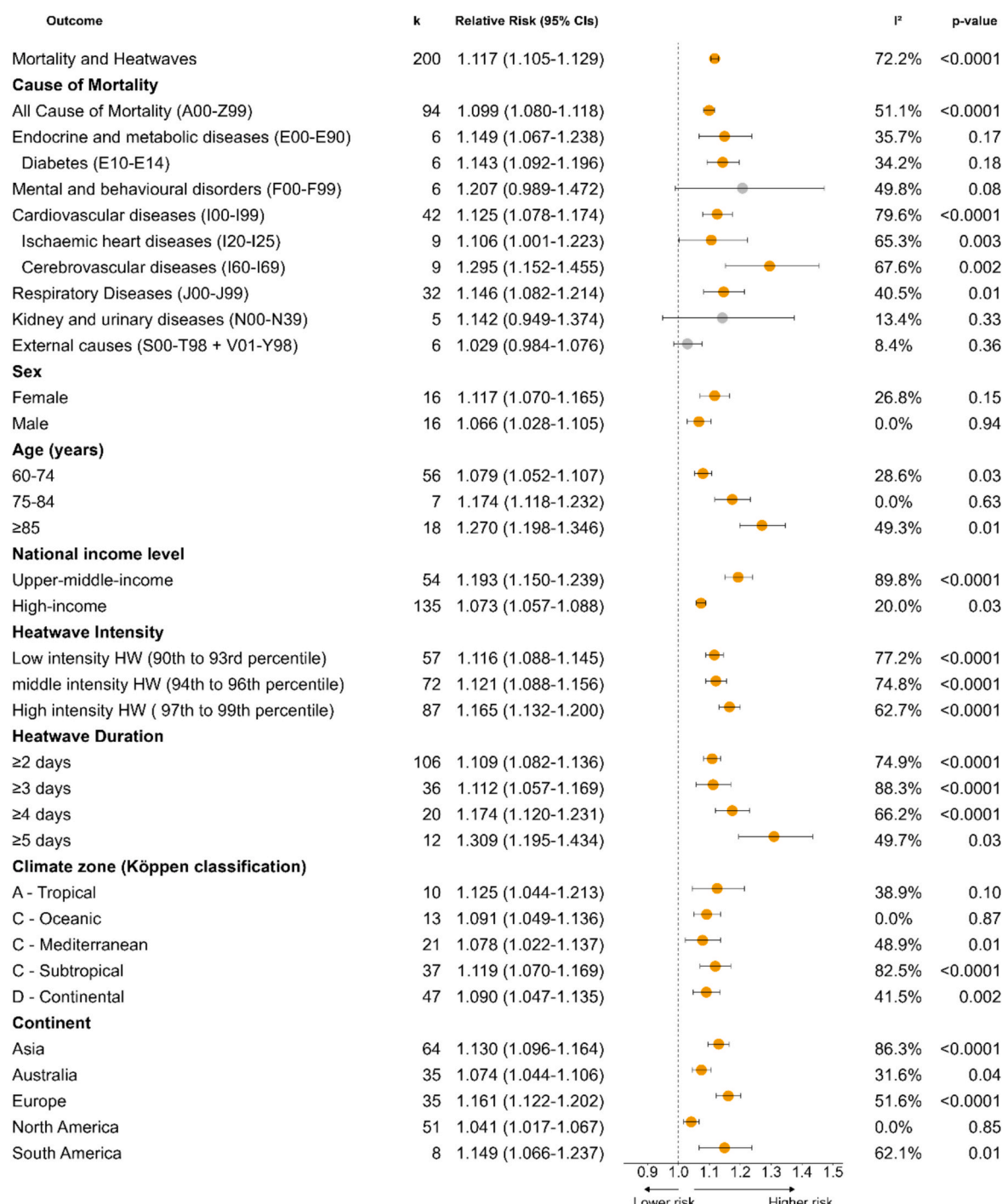


Fig. 7. Random-effects *meta*-analysis results showing change in RR and 95% CIs during heatwaves for mortality among older adults in different groups. k = the number of effects estimates; 95% CIs = 95% Confidence interval, I² = heterogeneity statistic; p-value = statistical significance of heterogeneity. Yellow markers indicate statistically significant associations; grey markers indicate non-significant associations.

assigned to the evidence, with potential of downgrading or upgrading based on the certainty factors outlined in Table 4.

The overall certainty of evidence for mortality and heat exposure was not downgraded for any factors. High temperatures had no observed heterogeneity (I² = 0%, p = 0.547) and heatwaves moderate heterogeneity (I² = 72.2%, p < 0.0001).

Morbidity studies were downgraded for inconsistency due to high heterogeneity, as indicated by both I² (high temperatures: I² = 84.1%, p < 0.0001, 80% PI (0.963–1.072); heatwaves: I² = 97.6%, p < 0.0001, 80% PI (0.766–1.845) and 80% prediction intervals which included unity. For these domains, downgrading reflected substantial

heterogeneity that remained unexplained after considering a wide range of a priori subgroup and meta-regression analyses (summarised in Appendix 1, Tables S9–S13), together with prediction intervals indicating that future studies could plausibly include either changes in the magnitude of the effect estimates or null effects.

Evidence certainty for heat exposure for mortality and morbidity was upgraded for showing an exposure–response gradient.

Heat exposure and healthcare costs were downgraded for inconsistency and imprecision due to the limited available studies, reflecting the small number of estimates and the resulting statistical uncertainty in the pooled effects. Although both studies showed a dose–response

Table 3
Summary of included studies on health care costs of heatwaves and high temperatures.

Study	Location	Study Period	Study Design & Statistical methods	Exposure	Outcome	Age	Main Findings
Lin et al. 2012	New York State, USA	1991–2004	TS, Generalized Additive Model with Poisson distribution	Apparent temperature, per 1°F ($\approx 0.56^\circ\text{C}$) degree increase > 90°F (32.2°C)	Respiratory hospitalisations; direct inpatient costs	65–74, ≥ 75	Age 65–74 years: RR 1.16, 24 excess admissions, US\$ 199,609 in excess hospitalisation costs per year. Age ≥ 75 years: RR 1.17, 27 excess admissions, US\$ 236,496 in excess hospitalisation costs per year.
Wondmagegn et al. 2021	Adelaide, Australia	2014–2017	TS, Generalized Linear Model combined with a Distributed Lag Non-linear Model	Heatwaves, Excess Heat Factor	All-cause emergency department (ED) visits; direct medical ED costs	≥ 65	Low-intensity heatwaves (EHF = 0.01–37.4 K ²): RR 1.08 (95% CI 1.02–1.14); AU\$ 242,800 (95% eCI 37,000–440,000). Severe heatwaves (EHF = 37.5–98.6 K ²): RR 1.31 (95% CI 1.05–1.62); AU\$ 289,300 (95% eCI 101,600–484,100). All heatwaves (EHF = 0.01–98.6 K ²): Total attributable ED cost = AU\$ 530,100 (95% eCI 160,200–890,400).
Yang et al. 2024	Nanjing, China	2013–2017	TS, Distributed Lag Non-linear Model with Poisson regression	Mean temperature per 1°C increase	Hospitalisations and medical costs for ureter calculus	≥ 65	Increased daily medical costs for ureteral calculi RR 1.06 (95% CI 0.97–1.17); total medical costs for older adults were 48.26 million CNY, with an average cost per admission of 15.62 ± 15.77 thousand CNY.
Wang et al. 2021	17 sites, China	2014–2018	TS, Generalized Additive Model with Poisson regression	Mean temperature, 95th vs 50th percentile temperature	All-cause ED visits; direct medical costs (examination + medication)	60–74, ≥ 75	Examination costs: RR 1.01 (95% CI 0.92–1.12) for ages 60–74; RR 0.90 (95% CI 0.77–1.06) for ≥ 75 . Medication costs: RR 0.97 (95% CI 0.88–1.08) for ages 60–74; RR 1.04 (95% CI 0.90–1.21) for ≥ 75 . National ED visits costs (not age specific) 274 million CNY (95% CI 95–437 million).

Table 4
Summary of the certainty and strength ratings of the body of evidence on the influence of high temperatures and heatwaves on mortality, morbidity, and healthcare costs.

	High temperature studies			Heatwave studies		
	Morbidity	Mortality	Health Care Costs	Morbidity	Mortality	Health Care Costs
Initial rating	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate
Certainty of evidence assessment						
Down grading factors						
Risk of bias across studies	0	0	0	0	0	0
Indirectness	0	0	0	0	0	0
Inconsistency	–1	0	–1	–1	0	–1
Imprecision	0	0	–1	0	0	–1
Publication Bias	0	0	0	0	0	0
Upgrading factors						
Large magnitude of effect	0	0	0	0	0	0
Dose response	+1	+1	0	+1	+1	0
Confounding minimises effect estimates	0	0	0	0	0	0
Summary of the certainty assessment	Moderate	High	Low	Moderate	High	Low

relationship, no upgrading occurred due to the small number of available studies.

No exposure–outcome domain was upgraded for “large magnitude of effect,” as pooled relative risks for high temperatures and heatwaves were consistently below conventional Navigation Guide/GRADE thresholds for very large effects.

The overall certainty of evidence was rated high for mortality, moderate for morbidity, and low for healthcare costs (Table 4). Leave-one-out sensitivity analysis showed no significant changes to the pooled estimates for heat exposure and morbidity or mortality outcomes. Sensitivity analyses demonstrated consistency in both the direction and magnitude of the associations across the studies (Appendix 1, Tables S11–S14).

The overall strength of evidence was rated as “sufficient” for the association between heat exposure and increased both mortality and morbidity among older people, based on numerous well-designed

studies demonstrating a positive association, with reasonable confidence that chance, bias, and confounding factors have been ruled out as explanations. It is unlikely that future research will significantly alter this conclusion to null or statistically insignificant.

For healthcare costs and heat exposure, the evidence was rated as “limited” due to reliance on the limited number of studies, meaning future research could significantly alter these findings. (See Table 4)

4. Discussion

4.1. Key findings

To the best of our knowledge, this systematic review provides the largest and most comprehensive overview to date of the global epidemiological evidence on heat exposure and its health effects on older adults. Encompassing 623 studies from six continents and 305 locations

classified according to the Köppen-Geiger, this review provides an updated synthesis and quantifies mortality and morbidity effects from heat exposure across different climate zones, demographic factors, and income classifications, while offering a narrative synthesis of healthcare costs due to limited available data. Most importantly, it identifies and quantifies disease-specific mortality and morbidity outcomes for older individuals. This review also stratified outcomes by sex and within older age groups (e.g., 60–74, 75–84, and ≥ 85 years), to better understand who is most at risk.

Morbidity increased by 1.6% (95% CI: 1.2–1.9) per 1°C rise in temperature and by 18.9% (95% CI: 14.6–23.3) during heatwaves, with notable elevations in heat-related illnesses and external causes for both exposures. Mortality rose by 2.2% (95% CI: 2–2.4) per 1°C increase and by 11.7% (95% CI: 10.5–12.9) during heatwaves. The leading causes of excess risk differed by exposure but were highest for external causes including heat-related illnesses and intentional self-harm, as well as cerebrovascular and endocrine diseases.

Only four studies quantified the economic burden of heat on healthcare systems, showing significant cost increases for emergency department visits, respiratory admissions and ureter calculus hospitalisations (Wondmagegn et al., 2021, Lin et al., 2012, Yang et al., 2024, Wang et al., 2021).

4.2. Context with previous evidence

The health impacts of extreme heat on older adults are widely recognised, yet age-specific quantitative syntheses remain scarce. Only few systematic reviews have specifically concentrated on older adults, providing quantitative analyses of heat-related disease and mortality outcomes. Bunker et al. undertaken in 2016, is the only systematic review to date that *meta*-analysed 37 studies to quantify cause-specific heat effects on both mortality and morbidity across multiple disease categories in older populations. They found that per 1°C increase in temperature, mortality increased by 3.44% (95% CI: 3.10–3.78) for cardiovascular, 3.60% (95% CI: 3.18–4.02) for respiratory, and 1.40% (95% CI: 0.06–2.75) for cerebrovascular diseases. For morbidity, the largest increases were observed for heat-related illnesses (14.83%, 95% CI: 8.22–21.84), respiratory diseases (2.76%, 95% CI: 1.51–4.03), diabetes (1.02%, 95% CI: 0.43–1.62), and acute renal failure (2.12%, 95% CI: 1.59–2.65). However, the effects of heatwaves were not investigated, nor were risk of bias evaluations or climatic and geographic variability addressed.

A recent systematic review and *meta*-analysis by Byun et al. (2024) synthesised evidence on high temperatures and heatwaves in relation to mental and neurological outcomes, including studies with older populations. Their *meta*-analysis of five studies found that each 1°C increase in temperature was associated with an increase in hospital visits for mental disorders (RR: 1.014, 95% CI: 1.001–1.026). For heatwaves, the pooled effect estimates showed a rise in hospital visits (RR: 1.269, 95% CI: 1.030–1.564), while mortality from mental disorders showed a similar pattern (RR: 1.266, 95% CI: 0.956–1.678), although this was not statistically significant. The review also identified substantial methodological heterogeneity across studies, especially regarding heatwave definitions and temperature thresholds.

Furthermore, a 2012 *meta*-analysis estimated a 2–5% rise in all-cause mortality per 1°C among those ≥ 65 years (Yu et al., 2012) and narrative reviews have reported consistent heat-wave-related increases in cardiovascular and respiratory mortality (Åström et al., 2011, Kaltsatou A, 2018).

Our updated findings are broadly consistent with previous systematic reviews and show similar magnitudes for key outcomes in older populations, particularly for cardiovascular, respiratory, and cerebrovascular conditions. Notably, we also identified additional cause-specific risks, including e.g. external causes, intentional self-harm, digestive diseases, COPD, asthma, and dementia/Alzheimer's disease, that were not previously synthesised. These likely reflect both the

expansion in the available evidence and a growing recognition of non-traditional heat-related health outcomes.

Overall, our pooled estimate of a 2.2% (95% CI: 2–2.4) increase in mortality per 1°C rise in temperature falls within the 2–5% range reported by Yu et al (2012).

These differences in outcomes may be explained by the substantially larger number of included studies, many using recent multi-city datasets with adaptive threshold modelling and the inclusion of a broader, more up-to-date evidence base. This reflects the expanded scope of available research and captures a wider spectrum of heat-related morbidity and mortality, including more recent and previously underrepresented health outcomes.

Despite growing evidence on the health risks of heat, studies quantifying the economic burden remain sparse. A 2019 systematic review identified only 10 studies on heat-related healthcare costs, with just one focusing specifically on older adults (Wondmagegn et al., 2019). Reported costs included ambulance services, emergency department visits, hospital admissions, medical devices, and diagnostic procedures (Wondmagegn et al., 2021, Lin et al., 2012, Yang et al., 2024, Wang et al., 2021). These data suggest that the economic impact of extreme heat on healthcare systems is substantial yet under-investigated in older populations.

Our review identified only three studies quantifying healthcare costs in the context of heat, all reporting significant increases in expenses related to respiratory admissions and emergency visits. However, there is still no comprehensive synthesis that quantifies heat-attributable costs by disease type, age group, or climate zone. This underscores the need for more targeted economic evaluations, particularly as the ageing population continues to grow and climate-related health risks intensify.

4.3. Subgroup analysis and potential underlying mechanism

External causes of morbidity, particularly heat-related illnesses such as heatstroke, heat exhaustion, and dehydration, significantly increased during heatwaves, confirming previous findings (Faurie et al., 2022, Bunker et al., 2016). Prolonged exposure to high temperatures can overwhelm the body's ability to cool itself, leading to fluid and salt loss and symptoms like nausea, confusion, and seizures. If untreated, this can result in multi-organ failure (Hifumi et al., 2018). Older adults, with diminished thirst perception and widespread use of medications like diuretics, are at greater risk of dehydration (Meade et al., 2020).

CVDs, particularly cerebrovascular conditions, were strongly associated with heat exposure and mortality. This finding was in line with previous studies (Bunker et al., 2016, Liu et al., 2022, Cheng et al., 2019) and could be explained by heat exposure increasing skin blood flow and sweating, leading to volume depletion, sympathetic nervous system activation, and elevated heart rate (Crandall and Wilson, 2015). This is particularly challenging for older adults, as their physiological thermoregulatory capacity decreases with age. Specifically, sweat gland output becomes less effective, blood flow to the skin is reduced, and increases in cardiac output are smaller (Meade et al., 2020). This raises metabolic demand, potentially causing ischemia or thrombosis (Peters and Schneider, 2021). Fluid loss and electrolyte imbalances further increase the risk of arrhythmias and myocardial ischemia (Liu et al., 2015). Older people are also more likely to take medications that can impair thermoregulatory responses such as antihypertensives, or diuretics that increase water excretion and thus predispose to dehydration. These medications are linked to increased mortality during higher temperatures (Ebi et al., 2021).

The higher CVD mortality compared to morbidity may result from several factors, including the possibility that patients with severe CVD die before receiving medical care. Limited access to healthcare and cases that do not result in hospitalisation or emergency department visits can contribute to underreporting of morbidity, especially during heat periods (Lee et al., 2024, Michelozzi et al., 2009). Additionally, older adults face heightened risk due to multiple health conditions, increased

physiological fragility, and difficulties accessing care on their own (Lee et al., 2024, Leon and Helwig, 2010). To better capture the effects of temperature on morbidity, data from outpatient settings, such as general practices and home visits, should be included, as they are currently underrepresented (Bunker et al., 2016, Smith et al., 2016).

Respiratory mortality and morbidity were significantly elevated, consistent with previous studies (Achebak et al., 2024, Bunker et al., 2016). The mechanisms leading to adverse respiratory outcomes are not fully understood but are likely influenced by thermoregulatory pathways (Michelozzi et al., 2009, Achebak et al., 2024). Heat exposure can worsen conditions like COPD by increasing ventilation and inflammatory responses (Viegi et al., 2007, Anderson et al., 2013, White, 2006, Leon and Helwig, 2010). COPD is a common medical condition among older people (Bunker et al., 2016, Mollica et al., 2021). This may lead to hyperventilation during extreme heat, increasing the risk of hyperinflation, dyspnea, and cardiovascular complications (Sprung, 1980). Inhalation of hot air can also trigger adverse airway reactions (Anderson et al., 2013) and cause bronchoconstriction in patients with asthma (Hayes et al., 2012). However, this study found a significant effect for COPD mortality per 1 degree increase but not for morbidity. No significant effects were found for asthma, or pneumonia, though the overall respiratory impact was notable across all exposures and outcomes.

Endocrine diseases increased with heat exposure, with diabetes showing significant associations with heatwaves and mortality. Previous reviews also found diabetes significantly elevated among older adults during heat exposure (Bunker et al., 2016, Song et al., 2021).

Diabetes is linked to various metabolic, cardiovascular, and neurological dysfunctions, which can also impact thermoregulatory mechanisms during heat exposure (Kenny et al., 2010). In both type 1 and type 2 diabetes, the blood vessels in the skin may have reduced ability to dilate, potentially leading to decreased blood flow to the skin's surface and impairing heat dissipation (Petrofsky et al., 2005, Stansberry et al., 1997, Wick et al., 2006).

Heat exposure is significantly related with kidney and urinary diseases mortality and morbidity. During further subgroup analysis, renal failure morbidity was observed to be an increased risk among older people. The kidneys play a key role in thermoregulation by regulating blood pressure and maintaining fluid and electrolyte balances (Robson, 2014). Dehydration and reduced blood flow during overheating increase the risk of kidney damage. Ageing kidneys take longer to correct imbalances under heat stress (de Lorenzo and Liaño, 2017, Rivas Velasquez et al., 2014).

Morbidity due to mental and behavioural disorders was significantly elevated during heatwaves, while no significant associations were observed for mortality from heatwaves or morbidity during periods of high temperature. Importantly, a notable increase in mortality from intentional self-harm during high temperatures was identified, further highlighting the mental health burden of extreme temperatures. These findings align with previous *meta*-analytic evidence reporting increased risks of heat-related morbidity and mortality due to mental health conditions among older adults (Liu et al., 2021b). Age-related factors, along with psychotropic medications, impair the body's ability to manage heat and increase the risk of confusion, delirium, and cognitive decline (Yezli, 2023, Maes et al., 2023). Heat exposure can also intensify psychological distress and trigger risk behaviours, including alcohol use, substance abuse, and suicidal tendencies, especially when individuals are unable to find relief from the heat (Liu et al., 2021b, Page et al., 2007).

While the pooled estimate for dementia and Alzheimer's disease during high temperatures was not statistically significant, the number of available estimates was limited. This may change as more studies emerge, given the high prevalence of these neurodegenerative conditions in older adults and their known vulnerability to heat.

People aged ≥ 85 years had the highest risk overall, but for heatwaves and morbidity, the 75–84 age group showed a dramatically higher risk (116.8%, 95% CI: 24.2–278.4) compared to those aged

60–64 (34.2%, 95% CI: 21.2–48.6) and ≥ 85 years (28.9%, 95% CI: 12.7–47.4). Nevertheless, this may change with future studies due to the limited sample size for this group ($n = 7$). There was little difference between males and females for high-temperature morbidity and mortality. However, for heatwaves and mortality, females showed a noticeably larger increase in mortality compared to males (11.7% (95% CI: 7–16.5) vs. 6.6% (95% CI: 2.8–10.5)) respectively. This may reflect both physiological and social differences, including reduced thermoregulatory capacity due to postmenopausal changes in estrogen levels and higher prevalence of living alone, lower income, and chronic health conditions among older women, all of which have been associated with increased heat vulnerability (van Steen et al., 2019). Heatwave-related morbidity could not be analysed due to insufficient data.

Climate change affects populations differently based on location. Most of the included studies were conducted in high- and upper-middle-income countries, with only two from a low-income country and a few from lower-middle-income countries. This imbalance may limit the generalisability of the findings to lower-income settings, which are more vulnerable to extreme heat due to limited resources and inadequate infrastructure, such as air conditioning, home insulation, and healthcare access (Bowen and Friel, 2012, Gamble et al., 2013).

The majority of studies in this *meta*-analysis were conducted in Climate Zone C (Mediterranean, Oceanic, Subtropical). Tropical regions had the highest morbidity risk from elevated temperatures, while Mediterranean climates recorded the highest mortality rates. Heatwave-related morbidity was most pronounced in subtropical climates, and heatwave-related mortality was highest in tropical climates. These findings indicate that mortality and morbidity risks vary across Köppen-Geiger climate classifications, potentially related to local climatic conditions or factors increasing vulnerability, such as socioeconomic status and healthcare infrastructure.

Furthermore, humidity is another key factor, as it exacerbates heat stress and may contribute to regional differences in heat-related health outcomes (Patel et al., 2019a, Longden, 2019). Sensitivity analyses showed virtually identical high-temperature effects on morbidity (no humidity adjustment: RR 1.019 [95% CI: 1.008–1.030] vs. adjusted: RR 1.015 [95% CI: 1.009–1.020]) and mortality (no adjustment: 1.028 [95% CI: 1.022–1.034] vs. adjusted: 1.021 [95% CI: 1.018–1.024]).

For heatwaves, morbidity risks were modestly higher in studies without humidity adjustment (RR 1.247 [95% CI: 1.097–1.418] vs. adjusted: 1.107 [95% CI: 1.072–1.144]), while mortality estimates remained similar (RR 1.053 [95% CI: 1.025–1.082] vs. adjusted: 1.134 [95% CI: 1.112–1.157]). These comparisons indicate that humidity adjustment or inclusion does not have a large impact on our overall temperature–health associations for morbidity and mortality (Appendix 1; Tables S11–S14). Nevertheless, humidity plays a critical role in amplifying physiological heat strain. Epidemiological studies may underestimate its effect due to methodological limitations, limited geographical coverage and data constraints, particularly the use of daily data and populations with impaired sweating responses. Moreover, at high temperatures, humidity may have limited additional influence on the health conditions that cause most heat-related deaths (Baldwin et al., 2023). Thus, the apparent similarity in effect estimates in our synthesis may reflect these analytical constraints rather than the absence of a physiological effect.

In addition, we explored heatwave intensity and duration gradients using subgroup *meta*-analyses. These analyses showed generally higher pooled risks for more intense and longer heatwave events for mortality, but effect estimates did not consistently increase across all intensity or duration categories, particularly for morbidity outcomes (Appendix 1, Tables S13 and S14). Accordingly, the presence of an exposure–response gradient was interpreted as supportive of an association, but not as evidence of a strictly monotonic increase in risk. Such patterns may be influenced by mortality and morbidity displacement, population adaptation, and differences in target effect estimands across studies.

Several factors contribute to the heightened vulnerability of older

adults to heat, including age-related physiological changes, comorbidities and associated medication utilisations, and environmental and social factors (Zhang et al., 2016, Cissé et al., 2022). Age-related impairments in thermoregulation, such as reduced sweat production and diminished blood flow to the skin, decrease the body's ability to dissipate heat, increasing the likelihood of cardiovascular complications. Furthermore, the combination of multiple chronic conditions and medications exacerbates the risk of adverse health outcomes in response to heat (Bunker et al., 2016, Meade et al., 2020). Other crucial factors that determine vulnerability are living alone, housing with inadequate cooling or heating, behavioural responses to heat stress, and poor access to health care (Bunker et al., 2016).

4.4. Uncertainty and heterogeneity in studies

Inter-study heterogeneity was generally high when assessing the relationship between morbidity and heat exposures, indicating inherent differences between the studies. Some subgroup analyses showed low to moderate heterogeneity, while others – such as disease groups, healthcare utilisation, study designs, and heatwave intensities – had higher heterogeneity. Despite adjustments for meteorological factors and air pollutants in most studies, variations in covariates, study populations, research methods, and contexts likely contributed to the heterogeneity. For mortality and heat exposures, heterogeneity was low to moderate. Subgroup and sensitivity analyses indicated consistency in the direction and magnitude of the effects, especially for high temperatures and mortality, which showed consistently low or no observed heterogeneity, indicating high consistency among studies. Factors like age structure, access to medical care, social services, housing quality, and air conditioning likely further amplified these differences in heterogeneity (Bunker et al., 2016). Additionally, morbidity studies can have greater variability due to a wider range of health outcomes, underreporting, and limited access to care, unlike mortality studies which often have more uniform endpoints. Observations that vulnerable people often die instead of being hospitalised during heat events may explain why heat impacts on morbidity appear smaller and more variable than on mortality (Bunker et al., 2016, Lee et al., 2024, Michelozzi et al., 2009, Turner et al., 2012).

Differences in study aims, effect definitions, and analytic choices imply that studies may target different underlying effects, making heterogeneity in effect estimates expected rather than indicative of low strength of evidence. Study-specific covariate structures could not be reweighted or standardised across all included studies, which may contribute to residual confounding and heterogeneity.

Finally, the absence of upgrading for “large magnitude of effect” reflects the conservative Navigation Guide/GRADE criterion, which reserves this domain for unusually strong relative effects rather than for modest but widespread risks. In the context of heat and heatwaves, relative risks well below two can still translate into large population-attributable burdens because a high proportion of the population is exposed. Global assessments have documented substantial and increasing burdens of heat-related mortality and morbidity associated with recent human-induced warming, illustrating how modest relative risks at the study level can generate large absolute impacts when exposure is ubiquitous (Vicedo-Cabrera et al., 2021, Zhou et al., 2024, Ebi et al., 2021, Romanello et al., 2021).

4.5. Strengths and limitations

This review included 623 studies, providing sufficient evidence to analyse a wide range of climate zones, income settings, and disease outcomes. This enabled detailed stratification by age, sex, climate classification, and income group. The limited number of additional studies identified through citation screening of existing reviews further supports the comprehensiveness of the original search strategy.

This study had several limitations. Only peer-reviewed journal

articles published in English were included. Significant publication bias and high heterogeneity were detected, indicating the possibility of unexamined factors.

Effect estimates were standardised to reflect per 1°C increases for comparability, but these increments are defined relative to each study's own metric and threshold, and the ranges and interpretations of these metrics (for example, dry-bulb temperature vs composite indices that incorporate humidity) could differ across studies.

Consequently, a 1°C increase in one metric is not physically equivalent to a 1°C increase in another and our pooled ‘per 1°C’ estimates should be interpreted as standardised statistical contrasts rather than strictly comparable physical exposure changes. In addition, standardisation was based on high-temperature thresholds defined in the original studies, not across the full exposure range, so the pooled estimates do not imply a fully linear exposure–response relationship.

For heat-wave studies, recent methodological work indicates that non-heat-wave temperatures can act as confounders and that the choice of temperature-adjustment strategy can substantially influence estimated heat-wave effects (Kim and Bell, 2024). In our review, most studies estimating heat-wave–health associations did not explicitly adjust for non-heat-wave-related temperatures; therefore, the pooled heat-wave estimates may plausibly underestimate the total health effects attributable to heat waves. This highlights the importance of careful analytical design and transparent reporting of temperature-adjustment strategies in future heat-wave epidemiological studies.

Estimates were also pooled across different disease groups to generate overall effects; however, cause-specific subgroup analyses were conducted to address variation in disease pathways and reporting. While many subgroups included a large number of estimates, others had fewer and should be interpreted with caution.

To avoid bias and over-representation of correlated estimates (e.g. multiple age groups or lag days within the same study), we summarised these using a fixed-effect model to obtain a single study-level estimate. This approach assumes that subgroup estimates reflect a common underlying effect within each study and provides an appropriate weighting of evidence without over-representing individual studies. While this may slightly smooth or attenuate subgroup-specific differences, it yields a more stable and interpretable pooled estimate (Riley et al., 2015, Borenstein, 2009, Deeks et al., 2024).

Given the large number of included studies, it was not feasible to assess every study-specific detail in the risk-of-bias evaluation. A consistent approach was applied across all studies, focusing on key domains such as exposure, outcome, and confounding. This ensured comparability but may have overlooked finer differences in exposure precision, outcome definitions, or covariate adjustment. However, the overall assessment captures the best available reflection of study quality based on reported information.

Despite including studies from various locations, there were limited numbers from low- and middle-income countries, tropical and polar regions, the African continent and South America. Studies from Africa were largely absent, while those from South America were limited to a few locations. These regions may experience greater vulnerability to the impacts of a warming climate among older populations, although vulnerability factors themselves were not directly assessed in this review. Additionally, most of the original studies lacked data on modifying factors such as medication use, pre-existing conditions, and comorbidities. Finally, this review included only four studies investigating heat-driven healthcare costs, limiting the strength of the evidence on this association.

5. Conclusion

Our findings underscore the urgent need for climate policies and tailored health interventions to mitigate the escalating morbidity and mortality risks among older adults. With global climate change expected to increase acute heat exposure and the growing vulnerability of an

ageing population, certain regions will face heightened demands on social and healthcare services, necessitating innovative policy solutions (Falchetta et al., 2024).

The review identified consistent associations between high temperatures, heatwaves and multiple heat-sensitive outcomes, with variation by exposure definition, outcome type, age group, sex, climate zone and national income classification. To better understand the observed variations across exposures, populations and regions, studies should also explore the underlying factors and local resources that influence vulnerability and adaptive capacity. This could include individual-level data on pre-existing conditions, medication use, heat-sensitive diseases and healthcare access. Additional studies are especially needed in underrepresented regions, including the African continent and South America, as well as polar and tropical climates, and low- and middle-income countries, to understand how health risks vary across populations.

Finally, few studies have quantified the healthcare-related economic burden of heat, despite its clear challenges on healthcare systems. Further cost-focused research is critical to inform policy and support the implementation of sustainable and equitable heat mitigation strategies.

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CRediT authorship contribution statement

Sabrina Günsche: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Matthew A. Borg:** Writing – review & editing, Validation, Methodology, Formal analysis, Supervision, Data curation, Conceptualization. **Olga Anikeeva:** Writing – review & editing, Validation, Supervision, Methodology, Formal analysis, Conceptualization. **Blesson M. Varghese:** Conceptualization, Data curation. **Yannan Li:** Validation, Data curation. **Dinesh Bhandari:** Validation, Data curation. **Jing Li:** Validation, Data curation. **Haoran Yang:** Validation, Data curation. **Yihan Shi:** Validation, Data curation. **Jingwen Liu:** Validation, Methodology, Data curation. **Keith Dear:** Writing – review & editing, Funding acquisition, Conceptualization. **Bertram Ostendorf:** Writing – review & editing, Funding acquisition, Conceptualization. **Nigel Stocks:** Writing – review & editing, Funding acquisition, Conceptualization. **Danny Liew:** Writing – review & editing, Funding acquisition, Conceptualization. **Peng Bi:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envint.2026.110129>.

Data availability

Data will be made available on request.

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